Inverted MAML Card 1: Round 1 Thermal Bit-Slicing Report

Hardware: Card 1 (T120+T20, 1k/2.2k) | **FW:** v2.2-MAML (0x20261030) | **Params:** 40% PWM, K=1, LR=0.03, L2=1024, Mom=0.9, Clamp=±75 LSB

1. Round 1 Summary & Cumulative Comparison (R1–R1)

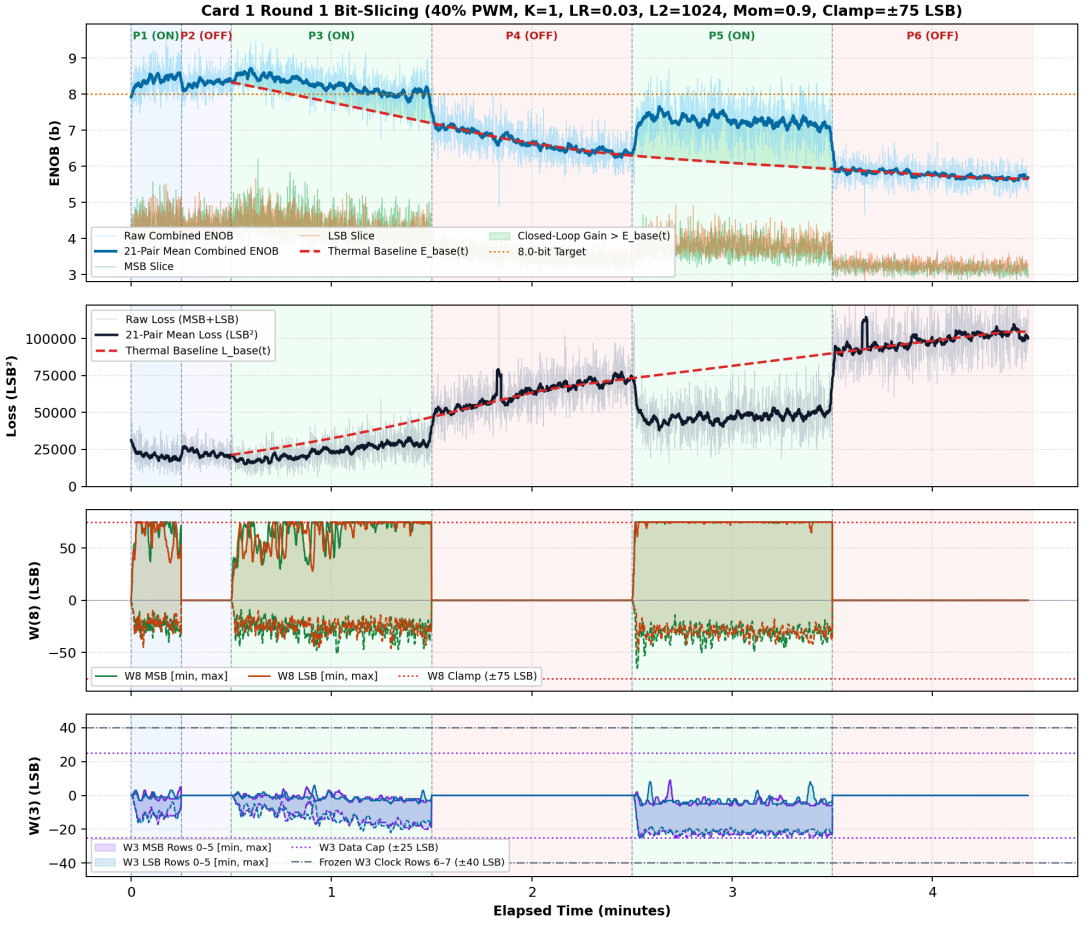
• **Cold Verification:** Phase 1 ON = **9.45 b**, Phase 2 OFF = **9.45 b** (W8=0 strictly).

• **Round 1 Detrended Trend Metrics:** Lift vs E_base = **+1.00 b**, Loss Ratio = **1.71x**, Climb = **+0.95 b**, Jitter = **0.40 b**, Recovery = **0.00**, Score = **+1.799**.

Rnd	LR	L2	Mom	Clamp	Lift	LossR	Climb	Jitt	Recov	Score
R1dc_fix*	0.03	1024	0.9	±75	+1.00b	1.71x	+0.95b	0.40b	0.00	+1.799

2. Phase-by-Phase Telemetry (4,216 Pairs)

Phase	Regime	N	Start->Late (Mean)	Δ vs E_base	Loss Start->Late	W8 Pk [M,L]	W3 Pk	W8 nz
P1	Cold / Regler ON	330	8.01->8.48 (8.38b)	-	29219->18810	[75,75]	17	330
P2	Cold / Regler OFF	309	8.10->8.35 (8.30b)	-	27016->21048	[0,0]	0	0
P3	Warm / Regler ON	1062	8.32->8.02 (8.25b)	+0.61 b	20693->28567	[75,75]	22	1062
P4	Warm / Regler OFF	920	7.23->6.42 (6.69b)	+0.02 b	48251->70589	[0,0]	0	0
P5	Warm / Regler ON	839	6.56->7.18 (7.25b)	+1.20 b	69339->48933	[75,75]	25	839
P6	Warm / Regler OFF	756	5.88->5.68 (5.77b)	+0.01 b	94044->103189	[0,0]	0	0



3. All W(8) (7x6 MSB+LSB) & W(3) (8x6 MSB+LSB) Weight Dynamics — Round 1

• **Full 180-Synapse Per-Pair Capture:** Every individual weight in **Matrix 8 MSB (42 weights)**, **Matrix 8 LSB (42 weights)**, **Matrix 3 MSB (36 active + 12 clock)**, and **Matrix 3 LSB (36 active + 12 clock)** is plotted below across all columns `Col 0..5`. W(8) enforces anti-windup clamping at ±75 LSB, while W(3) rows 0–5 stay within ±25 LSB so rows 6–7 clock staircases remain locked at ±40 LSB.

Round 1 Full Hardware Weight Telemetry: ALL 84 W(8) [7x6 MSB+LSB] & ALL 96 W(3) [8x6 MSB+LSB] Synapses

